

# CSE 280 Challenge Set 04 - Solutions

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## Question 1

For each of these functions, determine if they are well-defined, one-to-one, onto, and/or a bijection. For these problems, let  $A = \{x \mid x \in \mathbf{Z}, 0 < x < 5\}$  and let  $B = \{x \mid x \in \mathbf{Z}, 0 \leq x \leq 5\}$ . You may want to draw a digraph of each function (with dots and arrows). Recall that if not well-defined, then it is neither 1-1, onto, nor a bijection.

| Function                                                                               | Well Defined | 1-1 | Onto | Bijection |
|----------------------------------------------------------------------------------------|--------------|-----|------|-----------|
| $f : A \rightarrow A$ , where $f = \{(1, 2), (2, 2), (3, 3), (4, 3)\}$                 | x            |     |      |           |
| $f : A \rightarrow A$ , where $f = \{(1, 1), (2, 2), (3, 4), (4, 3)\}$                 | x            | x   | x    | x         |
| $f : A \rightarrow A$ , where $f = \{(2, 2), (3, 4), (4, 3)\}$                         |              |     |      |           |
| $f : A \rightarrow B$ , where $f = \{(1, 2), (2, 3), (3, 4), (4, 5)\}$                 | x            | x   |      |           |
| $f : B \rightarrow A$ , where $f = \{(0, 1), (1, 1), (2, 2), (3, 2), (4, 4), (5, 3)\}$ | x            |     | x    |           |

## Question 2

For each of these functions, determine if they are well-defined, one-to-one, onto, and/or a bijection. Consider using a graphing calculator like <https://www.desmos.com>.

| Function                                                                                     | Well Defined | 1-1 | Onto | Bijection |
|----------------------------------------------------------------------------------------------|--------------|-----|------|-----------|
| $f : \mathbf{Z} \rightarrow \mathbf{Z}$ , where $f(x) = 5x - 7$                              | x            | x   |      |           |
| $f : \mathbf{Z}^+ \rightarrow \mathbf{Z}^+$ , where $f(x) = \lceil \frac{x}{5} \rceil$       | x            |     | x    |           |
| $f : \mathbf{R} \rightarrow \mathbf{R}$ , where $f(x) = x^5 - 3x$                            | x            |     | x    |           |
| $f : \mathbf{R}^+ \rightarrow \mathbf{R}$ , where $f(x) = \frac{1}{x}$                       | x            | x   |      |           |
| $f : \mathbf{R} \rightarrow \mathbf{R}$ , where $f(x) = \log_5(x^2)$                         |              |     |      |           |
| $f : \{x \mid x \in \mathbf{R}, x > 0\} \rightarrow \mathbf{R}$ , where $f(x) = \log_5(x^2)$ | x            | x   | x    | x         |

| Function                                                                                              | Well Defined | 1-1 | Onto | Bijection |
|-------------------------------------------------------------------------------------------------------|--------------|-----|------|-----------|
| $f : \{x \mid x \in \mathbf{R}, x \neq 0\} \rightarrow \mathbf{R}, \text{ where } f(x) = \log_5(x^2)$ | x            |     | x    |           |

Answer:

- Not onto because you can't get every integer from an integer.
- Not 1-1 because multiple  $x$  map to the same value
- Not 1-1 because 3 different values of  $x$  map to the same  $y = 0$
- Not onto because missing 0 and negative
- Not well defined at  $x=0$
- All
- Not 1-1 because 2  $x$  map to the same value

## Question 3

### Part 1

Find the inverse  $f^{-1}(x)$  of the following functions:

- $f : \mathbf{R} \rightarrow \mathbf{R}$ , where  $f(x) = (x - 32) * 5/9$
- $f : \mathbf{R} \rightarrow \{x \mid x \in \mathbf{R}, x > 0\}$  where  $f(x) = \frac{1}{2^x}$
- $f : \mathbf{R} \rightarrow \mathbf{Z}$ , where  $f(x) = \lfloor x \rfloor$
- $f : \mathbf{R} \rightarrow \mathbf{R}$ , where  $f(x) = x^2$
- $f : \mathbf{R} \rightarrow \mathbf{R}$ , where  $f(x) = x^3$

Answer:

- $f^{-1}(x) = (\frac{9}{5} * x) + 32$
- $f^{-1}(x) = \log_2 \frac{1}{x}$
- Not a bijection (not 1-1). No inverse exists.
- Not a bijection (not 1-1 or onto). No inverse exists.
- $f^{-1} = \sqrt[3]{x}$

### Part 2

Select a value of  $x$  (per the domain) and calculate  $(f^{-1} \circ f)(x)$  for both functions in Part 1 to demonstrate that the inverse was successfully created.

- $f(41) = 5; f^{-1}(5) = 41$

- $f(2) = \frac{1}{4}; f^{-1}(\frac{1}{4}) = \log_2 4 = 2$
- N/A
- N/A
- $f(-5) = -125; f^{-1}(-125) = -5$

## Question 4

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Define the following two functions:

- $f : \mathbf{R} \rightarrow \mathbf{R}$ , where  $f(x) = \lceil x \rceil - x$
- $g : \mathbf{R} \rightarrow \mathbf{R}$ , where  $g(x) = x - \lfloor x \rfloor$

### Part 1

Graph function  $f$  using a graphing calculator. Explain what this so-called "sawtooth" function represents.

Answer:

$f$  represents the positive fractional distance from the ceiling  $[0, 1)$

### Part 2

Based on your answer above, does an inverse to  $f$  exist?

Answer:

No, an inverse does not exist because we can't determine the original real number from the fractional part.

### Part 3

Determine what  $f(x) + g(x)$  is equal to.

Answer:

$$f(x) + g(x) = \lceil x \rceil - \lfloor x \rfloor$$

- If  $x \in \mathbf{Z}$  then this is equal to 0
- Otherwise, this is equal to 1